A 51.19 mm², 4 TB/s bandwidth 3D Logic-on-SRAM Stacking CGRA Chip

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***Abstract*—CGRA delivers near-ASIC efficiency while retaining software-like programmability by leveraging spatially parallel PEs and distributed SRAM. However, in 2D realizations, bank-nonuniform, long compute–memory paths elevate wire delay and latency variance, tightening timing closure, and the SRAM capacity needed for throughput drives up silicon area and design cost. To address these limitations, we present a 28nm Logic-on-SRAM 3D-stacked CGRA chip that shortens memory access links and provides high-density vertical bandwidth via fine-pitch face-to-face hybrid bonding. The top logic die integrates 16 tile-aligned PE-array slices, each vertically paired with a 192-KB SRAM slice on the bottom die to realize near-memory access. We co-design the architecture and implementation using a holistic 3D flow that integrates multi-HB/V-HB co-P&R and ILM-based two-stage partitioning, with a unified cross-die signoff method encompassing timing, thermal, and SI. This enables robust 1 GHz/0.9 V operation. Post-silicon measurements demonstrate 4 TB/s die-to-die bandwidth (80.02 GB/s/mm²) and peak efficiency of 0.12 TFLOPS/W per slice. These results validate tile-aligned logic-on-SRAM stacking as an efficient and scalable path to higher compute density and local bandwidth in future CGRA design.**

Keywords—CGRA, 3D IC, Hybrid bonding

I. Introduction

The reconfigurable processing elements (PEs) and interconnects of a coarse-grained reconfigurable array (CGRA) enable rapid adaptation to diverse workloads (e.g., signal and image processing, wireless communications, and DNN acceleration), delivering near-ASIC computing power while preserving programmability. To boost the computing power, CGRA architectures usually employ a large number of PEs spatially distributed and inherently parallel, together with distributed SRAM blocks tightly coupled with PEs, rather than relying on a large centralized on-chip memory.

In conventional 2D CGRA implementations, scaling the PE array demands commensurately large SRAM cache arrays. The resulting macro footprints increase the physical distance between PEs and SRAM, lengthen global wires, and raise interconnect *RC* delay, which limits the attainable frequency and overall chip performance. Moreover, large SRAM arrays occupy substantial silicon area, exacerbating physical-design complexity (e.g., floorplan, power plan, clock tree synthesis, and routing) and, by inflating die size, significantly raising manufacturing cost. In addition, as illustrated in Fig. 1, the extensive SRAM footprint induces spatial variation in memory-access latency across the PE array, leading to inter-module timing imbalance and further complicating chip design. To overcome these limitations and enhance CGRA performance, we adopt 3D integration [1][2][3][4]. Vertical interconnect formed by TSVs and micro-bumps, together with fine-pitch face-to-face hybrid bonding (HB) [5][6][7], shortens PE–SRAM links and reduces interconnect delay. Moreover, vertical

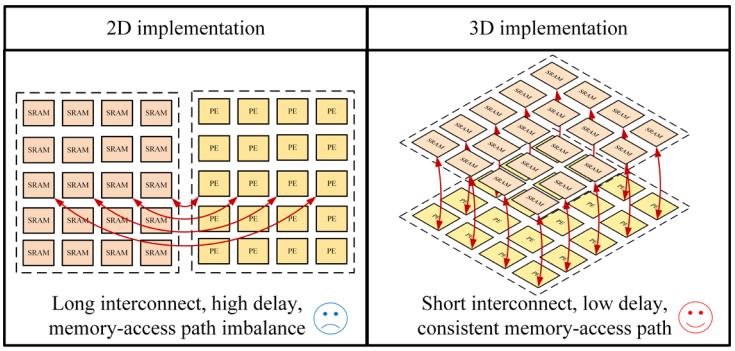


Fig. 1. Comparison between 2D CGRA and 3D CGRA.

stacking enables larger on-chip memory capacity and higher-density inter-die bandwidth directly beneath the compute fabric, thereby improving sustained throughput.

In this paper, we demonstrate a Logic-on-SRAM 3D-stacked CGRA SoC chip for the first time. Through co-optimization of the CGRA architecture, face-to-face HB technology, and a holistic design flow, we realize tile-aligned near-memory access with high 3D bandwidth and large memory capacity. Our silicon results establish a practical reference for scalable compute performance, memory capacity, and data bandwidth in future 3D CGRA chips.

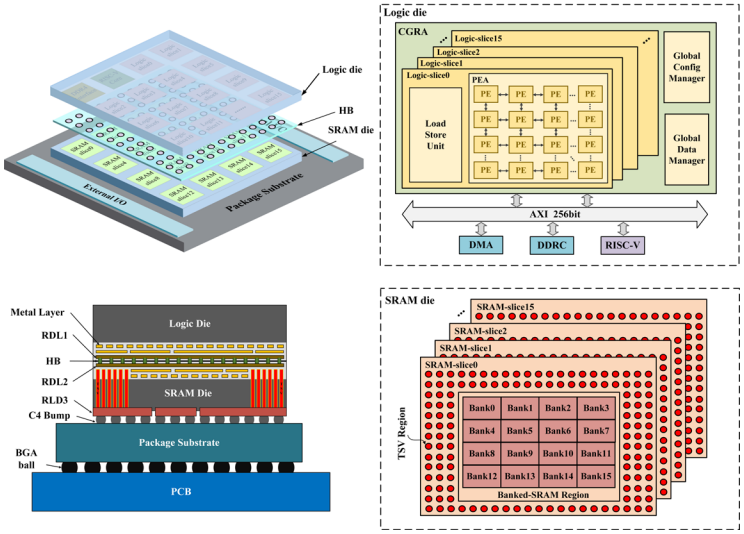


Fig. 2. The overall architecture of 3D Logic-on-SRAM stacking CGRA chip.

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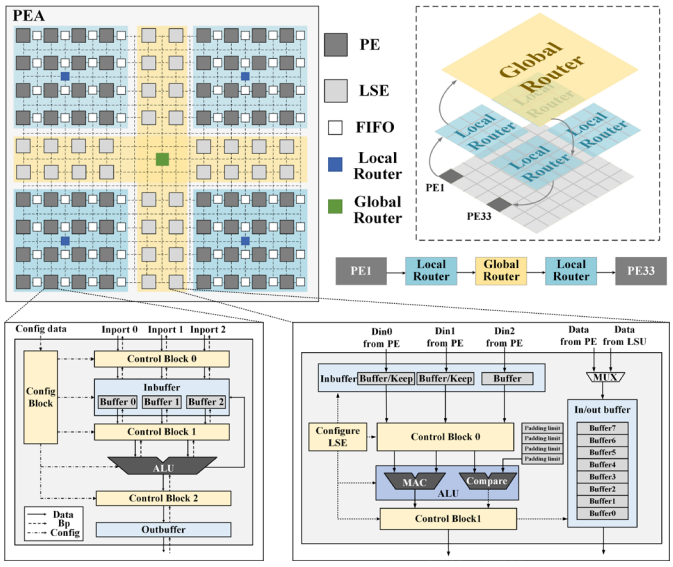


Fig. 3. The architecture of PEA, and communication between different PEs.

II. 3D Logic-on-SRAM Stacking CGRA Chip Architecture

Figure 2 illustrates the system architecture and hierarchical organization of the chip. The top die implements the CGRA logic layer, while the bottom die serves as the SRAM storage layer. This stacking configuration not only facilitates the insertion of TSVs between SRAM macros—thereby improving area utilization—but also positions the higher-power compute logic on the top die, enabling more efficient heat dissipation through external thermal management. The top logic die comprises a global configure manager, a global data manager, and sixteen reconfigurable logic slices. The bottom SRAM die contains sixteen corresponding SRAM slices. Each logic slice is vertically aligned and directly bonded to a 192 KB SRAM bank via HB, enabling tile-aligned near-memory access and higher effective data bandwidth.

The PE array (PEA) on the top logic die consists of PEs, load-store elements (LSEs), FIFOs, and routers as illustrated in Fig. 3. To support efficient management and communication, the 8×8 PE array is partitioned into four quadrants. Each quadrant operates in a spatially distributed manner with localized dataflow communication, thereby enhancing scalability in throughput, memory capacity, and data bandwidth. For efficient and flexible communication, data is transferred through a hierarchical routing structure. Intra-quadrant communication is managed by localized local router. For communication between quadrants, global routers provide an additional control layer, facilitating efficient and flexible cross-region coordination and data movement. To reduce critical path latency, data paths within PEs and LSEs are divided into multiple pipeline stages by input and output control blocks. An extra pipeline stage is also inserted prior to the SRAM interface to ultimately achieve a system frequency of 1 GHz. Additionally, each data path is equipped with bidirectional control signals, including 4-bit forward status flags and 1-bit backward backpressure signals for efficient and autonomous control flow handling.

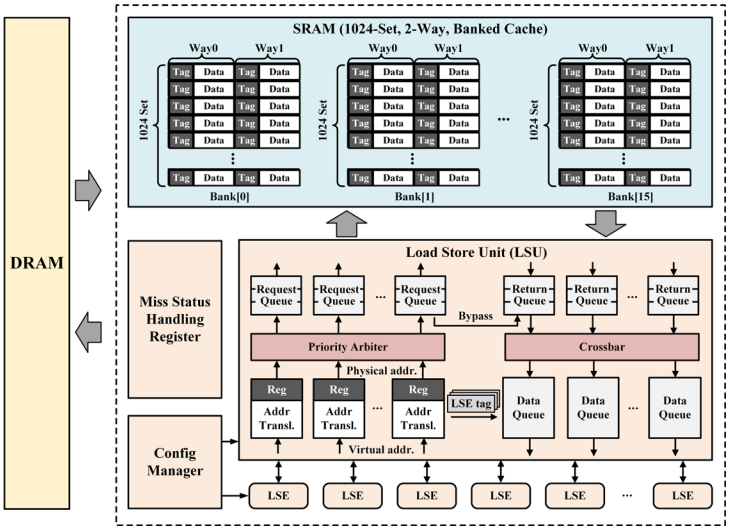


Fig. 4. The overall architecture of 3D memory access path.

To improve memory-access efficiency in the 3D chip, we design a multi-level hierarchical 3D memory access path comprising LSE-LSU-SRAM-DRAM, as shown in Fig. 4. To meet the memory access demands of diverse workloads, such as signal processing and DNN acceleration, the SRAM is organized as a 16-bank, 1024-set, 2-way set-associative cache structure. Each way contains a data array for storing payloads and a tag array for metadata, and supports concurrent 32-bit read and write operations. Consequently, each slice delivers 0.25 TB/s bandwidth, resulting in an aggregate chip-level bandwidth of 4 TB/s. The memory access logic, located on the logic die, includes the Miss Status Handling Register (MSHR), configure manager, Load-Store Unit (LSU), and LSEs. The access logic supports diverse memory access patterns, thereby meeting the generalized processing demands of the CGRA. Multiple LSEs can concurrently issue memory access requests to fetch as much data as possible, thereby keeping the PEs busy. The address translation unit inside the LSU maps the virtual address of LSEs to a physical address in the SRAM located on the bottom die. A priority arbiter then arbitrates among incoming requests and queues them for SRAM access. In the case of a cache miss, the MSHR handles the miss by fetching the required data from off-chip DRAM and updating the local cache. After data is retrieved from SRAM, it is placed into a return queue and routed through an internal crossbar switch, which distributes the data to the corresponding LSEs.

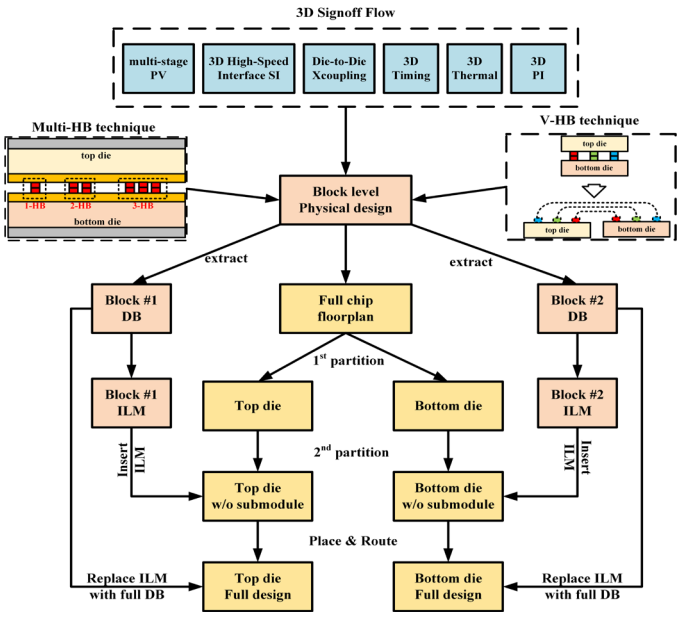


Fig. 5. The hierarchy physical design flow of 3D chip.

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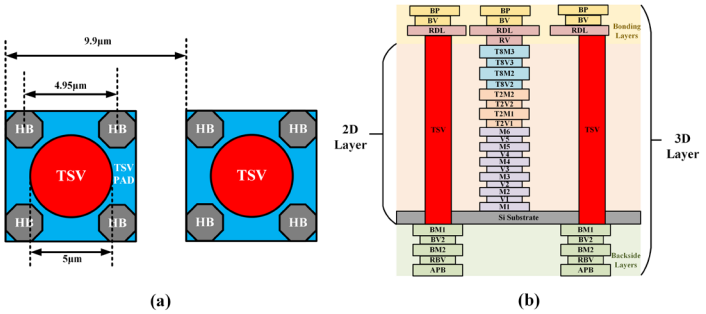


Fig. 6. 3D Process parameters. (a) TSV and HB process parameters. (b) 3D process structure.

III. 3D Chip Implementation Method

Signoff plays a vital role in chip design, and 3D signoff poses even greater challenges as it requires comprehensive analysis and validation across multiple domains. Immature design methodologies increase the difficulty of signoff and lengthen the design convergence cycle. Therefore, drawing on previous work [8], we integrate the 3D signoff flow with the 3D physical-design flow to form a unified, performance-driven 3D chip implementation methodology. Figure 5 presents the 3D chip implementation methodology, which combines submodule-level realization and top-level integration. At the submodule physical design phase, to support high-bandwidth 3D signaling, we adopt a multi-HB strategy in which multiple HBs are placed in parallel to carry one critical signal. This improves the robustness and signal integrity of the 3D interconnect. During place-and-route, we further introduce a virtual-HB (V-HB) abstraction that models 3D physical ports as zero-delay logical ports. This allows cross-die signal paths to be co-optimized under a unified timing view. After submodule closure, we extract interface logic models (ILMs) for both dies, retaining only input/output-related logic and timing views for use in top-level assembly. At the top-level physical-design phase, a two-stage partitioning flow is applied. In the first stage, the full chip is divided into a top die and a bottom die, in accordance with the functional constraints. In the second stage, large replicated modules within each die are replaced with their corresponding ILMs. This two-stage partitioning approach allows for substantial reductions in physical design complexity, which facilitates large-scale 3D integration. Upon completion of top-level routing, the ILMs are progressively swapped back with their corresponding full design databases to finalize full-chip 3D implementation.

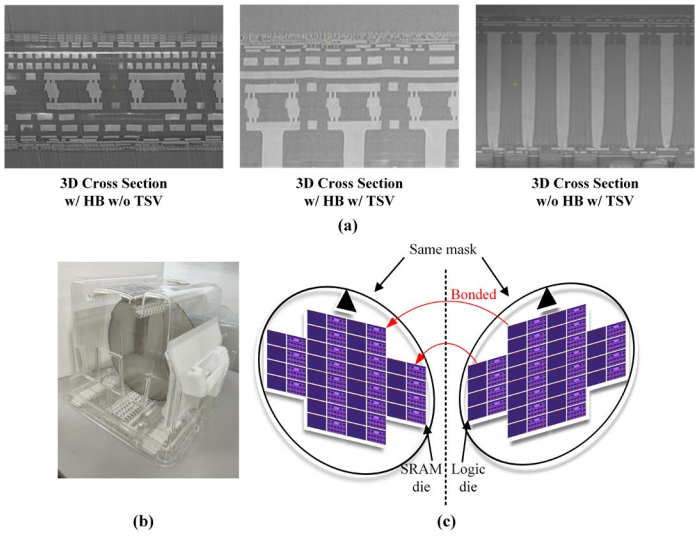


Fig.7. (a) Micrograph of 3D chip. (b) Photo of wafer. (c) Our wafer-on-wafer stacking method.

Table. 1. 3D chip performance under different workloads.

Workload	Performance		Bandwidth		Cache
	Activate PEs[a]	Performance (TOPS)	Activate LSEs[b]	3D bandwidth (TB/s)	Hit Rate
FFT	944	7.375	352	2.75	42.4%
MD	448	3.5	160	1.25	93.9%
NW	352	2.75	112	0.875	54.8%
BitonicSort	704	5.5	256	2	91.8%
OddEvenSort	432	3.375	512	4	51.7%
SPMV	416	3.25	496	3.875	77.8%
GEMM	960	7.5	384	3	88.4%
CONV	1024	8	368	2.875	86.0%

a. 1024 PE in total

b. 512 LSE in total

The unified flow is complemented by a comprehensive 3D signoff methodology that addresses timing, signal integrity (SI), power integrity (PI), thermal management, and physical verification (PV). For timing closure, we improve the 3D clock tree quality by constructing an integrated 3D clock tree. In addition to accounting for multiple corners and on-chip variations, we also include timing margins to accommodate wafer-to-wafer variations. SI challenges mainly arise from two aspects. As the two dies are bonded through HB, the reduced inter-die spacing increases die-to-die coupling capacitance and can degrade timing. To address this, we verify and optimize inter-die coupling by extracting the top three metal layers from both dies, performing parasitic extraction, and conducting timing analysis. Layout strategies such as avoiding long parallel interconnects across dies effectively mitigate coupling-induced delays and maintain timing integrity. Another signal integrity challenge arises from placing the compute die on the top layer, which increases the complexity of vertical signal paths, particularly for the DDR4 interface. To ensure DDR4 signal integrity, we conduct full-wave electromagnetic modeling and signal integrity simulations for the critical vertical channels. As a result, DDR4 operation at 2000MT/s is achieved with sufficient signal margin. Thermal simulations are conducted to evaluate temperature distribution across the chip under different workload conditions. When a single slice is active, the chip remains cool, while under full utilization with all sixteen slices operating, the peak temperature stays within safe operating limits. We optimize the IR drop of the top die by adjusting the number and placement of power TSVs, thereby improving power delivery integrity and timing robustness. For 3D PV, we first perform PV on the logic die and the SRAM die individually. Afterward, PV is conducted to ensure the 3D-specific layers, such as HB pads, frontside

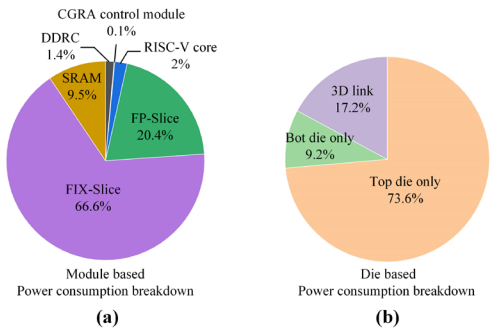


Fig. 8. Power consumption breakdown.

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Table. 2. Performance summary and comparison with prior 3D chip.

	This work	ISSCC 2022 [1]	ISSCC 2022 [2] 3D V-Cache	ISSCC 2024 [3]
Technology	28nm (Logic) 28nm (SRAM)	55nm (Logic) 25nm (DRAM)	7nm (Logic) 7nm (SRAM)	7nm (Logic+SRAM) 7nm (Logic+SRAM)
3D Stacking Method	Face-to-Face	Face-to-Face	Face-to-Back	Face-to-Face
3D Packaging Technology	Hybrid Bonding	Hybrid Bonding	Hybrid Bonding	Hybrid Bonding
3D bonding Pitch	4.95μm	3μm	9μm	2μm
Die area	51.19mm²	602mm²	81mm² (41mm² 3D SRAM)	15.17mm²
Total 3D Bandwidth	4TB/s	1.38TB/s	2TB/s	0.825TB/s
Bandwidth Density	80.02GB/s/mm²	2.35GB/s/mm²	49.95GB/s/mm²	55.69GB/s/mm²
Frequency	1GHz	300MHz	-	500MHz
Stacking Method	Logic-on-SRAM	DRAM-on-Logic	SRAM-on-Logic	(Logic+SRAM)-on-(Logic+SRAM)

RDL routing, and backside RDL routing, are properly designed, avoiding any misalignment during bonding. Finally, system-level PV is carried out on the fully integrated 3D stacked chip together with the package substrate, ensuring overall design integrity.

IV. Measurement Results

The logic and SRAM dies are both fabricated using a 28 nm CMOS process, with the corresponding layout shown in Fig. 7 and Fig. 9. To reduce mask costs, both the SRAM die and the logic die are integrated on the same wafer. A wafer-to-wafer 3D bonding approach is adopted, in which one of the wafers is flipped vertically (MY, 180°) and bonded to the other wafer using the same mask via HB. To facilitate this bonding, the HB pad layouts on both dies are designed symmetrically, and internal module placements are mirrored to simplify routing to HBs. Both dies share identical metal stack structures and mask configurations, enabling unified fabrication. To support flip-chip packaging and face-to-face stacking, the SRAM die is thinned to 50μm, while the logic die's I/O and power signals are first routed to TSVs and then completed via backside metal routing (RDL3). The 3D interconnect structure is illustrated in Fig. 6. The diameters of the TSVs and HBs are 5μm and 1.62μm, respectively, with pitches of 9.9μm for TSVs and 4.95μm for HBs. This 2:1 pitch ratio enables a cascaded connectivity scheme in which each TSV fans out to four HBs, such that each HB carries approximately one-quarter of the TSV current. This architecture improves overall current distribution and enhances system-level robustness and reliability.

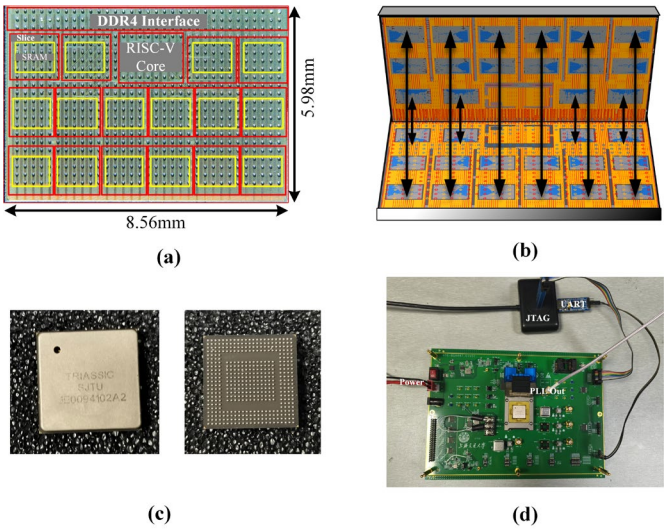


Fig. 9. Our fabricated 3D chip and evaluation platform. (a) 3D die photo. (b) 3D layout view. (c) Our fabricated chip. (d) 3D chip evaluation platform.

Table 1 and Fig. 8 summarize the chip's post-silicon measurement results. Table 1 reports sustained compute throughput (TOPS), effective 3-D bandwidth, and cache hit rate across different real workloads. Throughput scales with PE utilization: activating more PEs yields higher sustained TOPS, reaching 8 TOPS when all 1,024 PEs are enabled. Because the LSEs drive data movement, the effective 3D bandwidth tracks LSE utilization; for OddEvenSort workload, with 100% LSE utilization, the measured inter-die bandwidth attains 4 TB/s. Fig. 8 presents the power breakdown. At the module level, the power breakdown scales approximately with area. Since the logic die integrates twelve replicated fixed-point slices (FIX-Slice) and four floating-point slices (FP-Slice), the aggregate FIX-Slice power exceeds half of the chip total, whereas the FP-Slice aggregate is roughly one-third of the FIX-Slice total. The SRAM contributes only 9.5% of total chip power, attributable to its low dynamic power. In the die-level power breakdown, 3D-interconnect–related logic accounts for 17.2% of total chip power, whereas compute units on the upper logic die contribute 73.6%. Table 2 further benchmarks our 3D chip against other 3D designs, highlighting advantages from fine-pitch, face-to-face hybrid bonding and tile-aligned logic-on-SRAM stacking, which together enable high inter-die bandwidth and bandwidth density. Our 3D CGRA chip operates at 1Ghz with a 0.9 V supply. Through 3D stacking, we achieve a die-to-die bandwidth of 4 TB/s, corresponding to a bandwidth density of 80.02GB/s/mm². Our energy efficiency is 0.12TFLOPS/W/slice.

V. Conclusion

This work presents a 28 nm logic-on-SRAM 3D CGRA chip that stacks 16 tile-aligned PEA slices over per-slice SRAM via fine-pitch face-to-face hybrid bonding, enabling near-memory data paths and high local bandwidth. A co-designed 3D flow—multi-HB/V-HB co-P&R, ILM-based two-stage partitioning, and unified cross-die signoff—achieves robust 1 GHz/0.9 V operation. Post silicon measurements report 4 TB/s die-to-die bandwidth (80.02 GB/s/mm²) and 0.12 TFLOPS/W per slice. Together, the architecture, design method, and measurement results establish tile-aligned logic-on-SRAM stacking with fine-pitch HB as a practical, scalable foundation for CGRA.

VI. Acknowledgment

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